

Escuela Técnica Superior de
Ingeniería Informática
Planning And Progress Report Do4

Group C3.020 | Diseño y Pruebas II | 05/18/2025

Fecha	Versión	Autor
05-18-2025	1.0	Vacaro Goytia, Gabriel

Miembros:

- Student #5 Vacaro Goytia Gabriel gabvacgoy@alum.us.es

Repositorio de Github: <https://github.com/DP2-C1-068/Acme-ANS-Do4>

Contenido

Resumen ejecutivo 3

Introducción..... 4

Contenido..... 5

Conclusiones 22

Bibliografía 23

Resumen ejecutivo

En este documento se pretende englobar la planificación y progreso tanto individual como grupal del grupo C1.068, conformado por Gabriel Vacaro. Debido a que este reporte se está realizando en el transcurso del cuarto entregable, no va a ser posible reflejar las capturas del tablero de GitHub de los anteriores entregables, ya que simularlo no tendría mucho sentido a estas alturas. Sin embargo, se proporcionará un enfoque holístico y muy completo sobre todas las tareas realizadas en cuanto a la planificación se refiere y registros de progreso detallados.

Introducción

El presente informe tiene como objetivo recoger de forma estructurada y clara tanto la planificación como el progreso del grupo C1.068, conformado por Gabriel Vacaro, en el contexto de la entrega del proyecto completo. Este documento busca facilitar una visión integral del trabajo realizado, documentando las tareas llevadas a cabo, su planificación temporal, los costes asociados y los indicadores de rendimiento obtenidos a lo largo del desarrollo.

La planificación se presenta en detalle, con un listado de tareas clasificadas por roles, tiempos estimados y reales, así como un presupuesto desglosado. Aunque no se incluyen capturas del tablero de gestión de versiones anteriores por razones justificadas, se mantiene la coherencia metodológica exigida por la asignatura. En cuanto al progreso, se incluyen los registros individuales, con sus correspondientes evaluaciones, así como la gestión de conflictos, si los hubiera, y un análisis comparativo entre el presupuesto inicial y los costes reales.

En definitiva, este documento constituye un ejercicio de transparencia, autoevaluación y mejora continua que permite reflejar el compromiso, la organización y el esfuerzo del grupo en el desarrollo del proyecto.

Este informe se estructura de la siguiente forma: una portada inicial con la información del grupo, seguida de un índice de contenidos y un resumen ejecutivo. A continuación, se incluye una tabla de revisiones y la presente introducción. El cuerpo principal del documento se divide en dos capítulos: planificación y progreso. Finalmente, se presentan las conclusiones y la bibliografía.

Contenido

Planificación

Como este proyecto ha sido realizado por una sola persona, no existirá en la siguiente tabla una columna de Asignado, pues siempre será Gabriel Vacaro Goytia.

Tareas	Descripción	Tiempo estimado	Tiempo real	Roles
Task-000: Look for a group	Use the tools provided by the institution (forum), to analyse and select the best applicants for the project.	1h	3h	Manager
Task-001-Initialise Repository	Initialise the github organization and the repository, following the guidelines provided by lecturers	1h	30 min	Manager
Task-002: Create Planning Dashbaord and add D01 Tasks	up a GitHub project for each student (currently only student #5) and for the group to ensure clear tracking of project progress. Additionally, add the tasks for both related to deliverable D01	1h	1h	Manager
Task-003: Customise the starter project	Use the starter project "Hello-world-25.1.0" as a template and customize it in order to create the look and feel of Acme-ANS	2h	2h	Developer
Task-003 : Verify customized starter project works well/T1	Check that the Customized project ACME-ANS-D01 works smooth and it is ready for development	30 min	20 min	Developer
Group-Taks004: Produce a chartering report	Produce a chartering report in which all members must agree to commit to a goal and establish metrics to measure performance.	1h	1h 30 min	Manager
Task-004: Verify the chartering report /T1	Check the Document and look for errors.	10 min	15 min	Manager

Task-005: Provide a link to the GitHub planning dashboard	Provide a link to the GitHub planning dashboards, these link must be allocated in a txt into the folder ./reports of the project	5 min	5 min	Manager
Task-006: Modify the anonymous menu to include your favourite web. Student#5	Modify the menu.jsp file located in \src\main\resources\WEB-INF\views\fragments to add a link in the anonymous menu. The link should display your DNI followed by your surnames and name, and it should redirect to your favorite page.	20 min	20 min	Developer
Task-006: Verify that the anonymous menu includes Students#5 Favourite link/T1	Run the application and make sure that it works as intended. Go to Anonymous menú and look for Student#5 identification and press on it, check that it redirects to another web page correctly. Plus, go to a Follow up session and wait for the lecturer to validate the task	5 min	5 min	Developer
Modify the anonymous menu to include your favourite web Student#5/R1	The lecturer advice that the suboption with the identification of Student#5 was not showing, this was due an error with git and github. Waiting for new testing	10 min	10 min	Developer
Task-006: Verify that the anonymous menu includes Students#5 Favourite link/T2	check that the Anonymous menu displays the Student#5 identification and redirects to his favourite web page.	5 min	5 min	Developer
Task-007: Add Group and Student#5 tasks to dashboard for D02	Add the necessary group and Student#5 Tasks in the github project L01-S03-Working-together referred to D02 deliverable	1h	1h	Manager
Task-007: check the planning dashboard fits the guideline given on L01-S03/T1	Check the L01-S03 lesson and look if the planning dashboard fits the requirement	30 min	20 min	Manager
Task-007: Remake the planning dashboard following the feedback/R1	Take in count the feedback given from the lecturer and Remake the dashboard following the document L01-S03-Working-together.	2 h	3 h	Manager

Task-007: check the planning dashboard fits the guideline given on L01-S03/T2	Check the L01-S03 lecture and see if the planning dashboard meets the requirements.	10 min	10 min	Manager
Task-008: Update Framework to v.25.2.0 & project to D02	Follow the instructions in the USE platform video tutorial of Professor Rafael and update the framework and the project.	30 min	30 min	Developer
Task-019: Change the organization and document details from C1.059 to C1.068	Change all the details according to the old organization (C1.059) to the new one (C1.068)	10 min	15 min	Manager
Task-020: Single person group document	Prepare a document in ./reports folder with the objective of consolidating all pertinent messages regarding the special case of my one-person group.	1 h	1h 30 min	Manager
Task-020: Review the Single person group document/T1	Check the Single-person-case-document looking for errors.	10 min	10 min	Manager
Task-023: Forum post about boundary values	Make a post in the forum in order to get feedback about boundary values in all the mandatory requirements POST: https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id=89154_1&nav=discussionboard&conf_id=426211_1&forum_id=253522_1&message_id=460729_1	45 min	50 min	Developer
Task-009: Implement Technician Role. Student#5	Follow the instruction in lesson L02-S01-Models and queries and implement a new Role called Technician, this task is related to Requirement 3 of Student#5.	1 h	2 h 30 min	Developer
Task-010: Implement Maintenance Records on Aircrafts. Student#5	Follow the instruction in lesson L02-S01-Models and queries and implement a new Entity called Maintenance Records, this task is related to Requirement 4 of Student#5	2 h	3 h	Developer
Task-011: Implement Tasks done by technician on aircrafts. Student#5	Follow the instruction in lesson L02-S01-Models and queries and implement a new Entity called Task, this task is related to Requirement 5 of Student#5	2 h	2 h	Developer

Task-013: Implement Airport Entity. Group	Follow the instruction in lesson L02-S01-Models and queries and implement a new Entity called Airport, this task is related to Requirement 4 of Group	1 h	1 h	Developer
Task-014: Implement Airline Entity. Group	Follow the instruction in lesson L02-S01-Models and queries and implement a new Entity called Airline, this task is related to Requirement 5 of Group.	1h	1h	Developer
Task-015: Implement Aircraft Entity. Group	Follow the instruction in lesson L02-S01-Models and queries and implement a new Entity called Aircraft, this task is related to Requirement 6 of Group	1h	1h 30 min	Developer
Task-016: Implement Service Entity. Group	Follow the instruction in lesson L02-S01-Models and queries and implement a new Entity called Service, this task is related to Requirement 7 of Group.	1h	1h	Developer
Task-017: Implement Review Entity. Group	Follow the instructions in lesson L02-S01-Models and queries and implement a new Entity called Review, this task is related to Requirement 8 of Group.	1h	1h	Developer
Task-023: Create custom validation	As we have seen today in L02-S02, create custom validation for entity Aircraft, maintenance record, task and technician	5h	3h 30	Developer
Task-024: Ask in the forum for custom constraints in [D02-S05-4]	Ask for advice in the forum about custom constraints: https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id=89154_1&nav=discussion_board&conf_id=426211_1&forum_id=253522_1&message_id=460922_1	30 min	45 min	Developer
Task-025: Take the advise given in the forum and change the boundary values in attributes S05	Follow the indications given by professor rafael and set the explicit ranges of the following entities: • Technician • MaintenanceRecord • Task Also change the name of startedAt attribute in maintenanceRecord to a more client-related name. https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id=89154_1&nav=discussion_board&conf_id=426211_1&forum_id=253522_1&message_id=460339_1	2 h 30 min	3 h	Developer

Task-026: Delete unnecessary entities (airport, airline, service, review)	Delete unnecessary entities (airport, airline, service, review) that are not carefully implemented and are not requested by lecturers.	20 min	20 min	Developer
Task-018: Produce assorted sample data for informal testing. Group	Follow the instructions given on L02 lessons and produce assorted sample data to test your application informally, this task is related to Requirement 9 of Student#5	1h 40 min	1 h 10 min	Developer
Task-018: test if the data is valid and meticulously generated/T1	See if the aircraft.csv data is generated according to L02-S02 lesson.	20 min	20 min	Developer
Task-027: Ask in the forum for advice (moment and inspection dates) [D02-S03-6]	Ask in the forum for lecturer's advice to implement correctly the assorted sample data related to maintenanceRecord entity. Link: https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id=89154_1&nav=discussion_board&conf_id=426211_1&forum_id=253522_1&message_id=460922_1	40 min	35 min	Developer
Task012: Produce assorted sample data for informal testing. Student#5	Follow the instructions given on L02 lessons and produce assorted sample data to test your application informally, this task is related to Requirement 6 of Student#5	30 min	40 min	Developer
Task-012: Test if the data is meticulously generated and valid/T1	Look for singular values, missing values or errors in the database values	20 min	25 min	Developer
Task-009: Test Technician Role. Student#5/T1	Make the quality assurance informal testing and ask the lecturer for feedback in the next follow up.	1 h 40 min	2 h	Developer
Task-010: Test maintenance records/T1	Make the quiality assurance informal testing and ask the lecturer for feedback in the next follow up.	2 h	3 h	Developer
Task-011: Test task entity/T1	Make the quiality assurance informal testing and ask the lecturer for feedback in the next follow up.	2 h	2 h 30 min	Developer

Task-015: Test Aircraft Entity/T1	Make the quiality assurance informal testing and ask the lecturer for feedback in the next follow up.	1 h 15 min	1 h	Developer
Task-010: Fix maintenance record validation and moment restriction/R1	Reimplement some maintenance record restrictions on date attributes	1 h	1 h 15 min	Developer
Task-010: Test maintenance record entity/T2	Make informal testing for maintenance record entity and make sure it meets the requirements	1 h 30 min	1 h 20 min	Developer
Task-012: populate involves.csv following the works-for AcmeJobs example/R1	Take the advice given by lecturers and repopulate involves.csv.	30 min	30 min	Developer
Task-012: check if the assorted sample data is complete/T2	Make some informal testing in order to check that sample data suite is complete	30 min	40 min	Developer
Task-030: Prepare laptop for Follow up and visit CDC to test A4.32 like monitor	Prepare the necessary equipment to be able to make the follow-up as comfortable as possible.	1 h 30 min	1 h 20 min	Developer
Task-006: Ask Lecturer to review anonymous favourite link.	Ask for feedback about anonymous fav link feature	10 min	5 min	Developer
Task-004: Ask Lecturer to Review Chartering Report	Go to a follow-up session and check with the lecturer the chartering report.	20 min	5 min	Manager
Task-028: Ask the lecturer to review technician, maintenanceRecord and tasks implementation	Go to a follow up session and asks the lecturer if the implementation is valid, annotate all feedback recieved	30 min	35 min	Developer

Task-020: Ask Lecturer to review Single person case group document	Ask Lecturer to review Single person case group document and annotate his feedback	10 min	5 min	Manager
Task-029: Ask the lecturer to review the data generation of aircraft, technician, task and maintenanceRecord	Go to a follow up session and take lecturer feedback about population files	10 min	10 min	Developer
Task-031: Send D02 deliverable	Package and send Acme-ANS-D02	10 min	15 min	Operator
Task-031: Send D02 deliverable /T1	Test if the packaging is correct and no file is missing	20 min	15 min	Tester
Task-032: Prepare D03 project and database	Prepare the environment in order to work properly on D03 deliverable	20 min	10 min	Developer
Task-036: LicenseNumber incomplete requirement (initials must match) + populate technicians	Take the advice given in the forum and populate as well technician csv : https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253522_1&message_id= 461641_1 Also add a validation in technician license number: first letter must match the initial of the first name, and the second must match the initial of the surname	30 min	40 min	Developer
Task-033: Update the framework to 21.3.0 for D03	Watch the video posted in EV and follow the steps to update the framework (in both workstations)	50 min	1 h 30 min	Developer
Task-035: Test listing and show Tasks /T1	Check if the implementation works and follow the indications given by the client	40 min	50 min	Developer
Task-034: Test listing and show Maintenance Records/T1	Check if the implementation works and follow the indications given by the client	30 min	45 min	Developer

Task-036 Ask in the forum for advice in relation to listing tasks and maintenance records	https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253522_1&message_id= 462839_1	1 h	40 min	Developer
Task-037: Test listing and show service for aircrafts/T1	Check if the implementation works and follow the indications given by the client	40 min	35 min	Developer
Task-035: Operations by technicians on tasks	Implement create, update, delete and publish services for tasks	6 h	8 h	Developer
Task-034: Operations by technicians on maintenance records	Implement create, update, delete and publish services for maintenance records	5 h	6 h	Developer
Task-037: Operations by administrations on aircrafts	Create and update aircrafts	2 h	2 h 30 min	Developer
Task-038: Create project Release for Prof corchuelo to test the framework for D04	Make a Github Release for lecturers in order to test the framework for next versions.	10 min	10 min	Developer
Task-035: Operations by technicians on Tasks/T1	Check the implementations of task features are complete and fulfill the requirements	3 h	4 h	Developer
Task-034: Operations by technicians on maintenance records/T1	Check the implementations of maintenance records features are complete and fulfill the requirements	3h	4 h	Developer
Task-037: Operations by administrations on aircrafts/T1	Check the implementations of aircraft features are complete and fulfill the requirements	1 h	1 h	Developer
Task-038: Check the detailed evaluation of the release/T1	Read and understand prof. corchuelo advice and evaluate the changes that must be done.	40 min	30 min	Developer

Task-038: Check the detailed evaluation of the release/R1	Make the suggested changes if pertinent	2 h 30 min	3 h	Developer
Task-038: Check the detailed evaluation of the release/T2	Check the implementations of the new changes make the features work as expected.	40 min	1 h	Developer
Task-040: Technician Dashboards	Implement technician dashboards	2 h	1 h 50 min	Developer
Task-040: Technician Dashboards/T1	Check the implementation works as expected and meets the requirements.	30 min	35 min	Developer
Task-041: Become a technician	Implement feature that makes the principal able to become a technician	1 h	55 min	Developer
Task-042: Update technician profile	Implement feature that makes the principal able to update his technician profile	1h 40 min	1 h 30 min	Developer
Task-041: Become a technician/T1	Check the implementation works as expected and meets the requirements.	10 min	10 min	Developer
Task-042: Update technician profile/T1	Check the implementation works as expected and meets the requirements.	40 min	35 min	Developer
Task-043: Operations on maintenance records by administrators	Make administrator able to list published maintenance records and list their tasks	1 h	30 min	Developer
Task-043: Operations on maintenance records by administrators/T1	Check the implementation works as expected and meets the requirements.	20 min	25 min	Developer
Task-044: Administrator Dashboard	Implement a dashboard for administrators in order to check the ratio of active and under maintenance aircrafts.	40 min	50 min	Developer
Task-044: Administrator Dashboard/T1	Check the implementation works as expected and meets the requirements.	5 min	5 min	Developer

Task-045: Group Supplementary I 21,22,23	Inform lecturers what did I know about WIS related topics.	30 min	40 min	Developer
Task-046: UML Diagram	Make a domain model diagram to ensure a better understanding of the requirements.	1 h 40	2 h 30 min	Analyst
Task-045: Group Supplementary I 21,22,23/T1	Check that reports are well structured and answer the questions succinctly.	10 min	10 min	Developer
Task-046: UML Diagram/T1	Check with prof. Corchuelo that UML diagram fits with the implementation that requires Acme-ANS.	10 min	10 min	Analyst
Task-048: Take the pieces of advice of Prof. Corchuelo and unauthorise some missing ilegal petitions	Take the advice given on follow-up implement them correctly.	40 min	50 min	Developer
Task-040: Technician Dashboards/R1	Fix “perCurrency” statistics to make it work as the client demands	1h	1 h 40 min	Developer
Task-049: Make informal testing before formal testing to ensure app is bug-free	Ensure that there are no hidden errors in Acme-ANS before formal testing.	3 h	2 h 30 min	Developer
Task-050: Formal testing for /technician/MaintenanceRecord and /technician/task features	Produce a test suite for maintenance record and task features.	7 h	5 h 40 min	Tester
Task-050: Formal testing for /technician/MaintenanceRecord and /technician/task features/T1	Check the test suite for maintenance record and task features. Look for coverage of instructions and data variations.	1h	1h 40 min	Tester

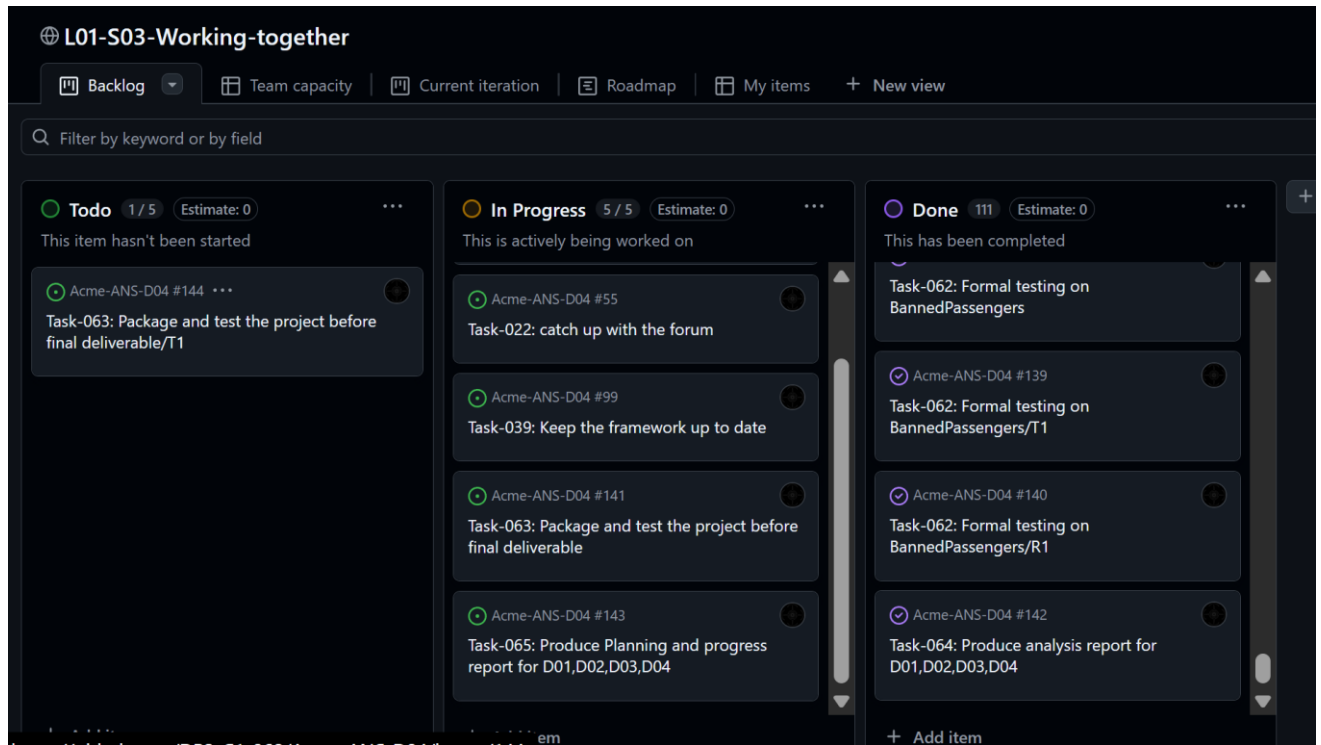
Task-050: Formal testing for /technician/MaintenanceRecord and /technician/task features/R1	Complete the test suite for maintenance record and task features with more test cases to improve data variations.	1 h	1 h 30 min	Tester
Task-051: Performance testing for /technician/MaintenanceRecord and /technician/task features	Collect data from <i>tester.trace</i> file and make the proper statistics according to L04 lessons.	3 h	4 h	Tester
Task-052: Produce a testing report	Follow the annexes document and produce a testing report.	3 h	2 h 50 min	Tester
Task-047: make a video in which individual requirements #8 and #9 are informally tested	Film a video with Microsoft Teams, upload it to OneDrive and share it with lecturers.	40 min	50 min	Developer
Task-050: Formal testing for /technician/MaintenanceRecord and /technician/task features/T2	Check the test suite for maintenance record and task features. Look for coverage of instructions and data variations.	2 h	1 h 30 min	Tester
Task-053: Perform ten mutations (5 individual + 5 group) and check test coverage	Perform 10 mutations, 5 are related to individual requirements and the other 5 are related to group requirements. For each mutation, provide evidence that test suite notices it.	2 h	2 h 10 min	Tester
Task-054: Create indices for aircraft entity	Adjust indices for aircraft entity according to existing database queries.	15 min	10 min	Developer
Task-055: Produce a Lint Report	Use sonar lint static code analyser and check for bad smells, evaluate if they are innocuous or not and provide justification.	1h 20 min	1 h 50 min	Developer

Task-051: Performance testing for /technician/MaintenanceRecord and /technician/task features/T1	Check the performance testing for maintenance record and task features (Z-test, p-value, 95% confidence intervals...).	40 min	20 min	Tester
Task-051: Fix some index that were unnecessary/R2	Index "Status" in maintenance record entity is unnecessary and must be deleted.	10 min	10 min	Developer
Task-056: Create entity Course.java	Look for a free external API that provide a list of courses and choose some attributes to implement course entity.	2 h	1 h 40 min	Developer
Task-057: Populate as administrator Course.java from external api	Create a controller to implement populate course service for administrators.	2 h 30 min	3 h	Developer
Task-058: List and show courses for technicians	Implement listing and showing courses by technicians	40 min	50 min	Developer
Task-059: Formal testing on Courses (populate, list, show)	Produce a test suite for entity Courses	30 min	30 min	Tester
Task-052: Produce a testing report/T1	Ask prof. Chorchuelo for feedback about the testing report.	10 min	10 min	Tester
Task-060: Create entity BannedPassenger.java	Create entity BannedPassenger	40 min	35 min	Developer
Task-061: List, show, create and update BannedPassengers	Make the controller, services and repository for listing, showing, creating and updating banned passengers by administrators	1 h	1 h	Developer
Task-062: Formal testing on BannedPassengers	Produce a test suite for entity BannedPassenger	2 h	1 h 50 min	Tester
Task-062: Formal testing on BannedPassengers/T1	Check the test suite for banned passengers feature. Look for coverage of instructions and data variations.	40 min	35 min	Tester

Task-062: Formal testing on BannedPassengers/R1	Complete the test suite for banned passengers feature with new test cases to cover entity changes.	1 h	1 h	Tester
Task-064: Produce analysis report for D01, D02, D03, D04	Produce a global analysis report for the whole course.	6 h	5 h 40 min	Analyst
Task-21: Attend theory lessons & Follow ups	Attend theory lessons in order to learn how to implement the requirements.	51 h 20 min	51 h 20 min	All roles
Task-022: catch up with the forum	Read all the unread messages in the forum and take notes from the feedback provided by lecturers, plus make my own posts	10 h	9 h	All roles
Task-039: Keep the framework up to date	Wait for new versions of the framework to come out and update it.	3 h	3 h	Developer
Task-065: Produce Planning and progress report for D01, D02, D03, D04	Produce a global planning and progress report for the whole course.	7 h	6 h	Manager
Task-063: Package and test the project before final deliverable	Make sure the project meets all the requirements and is ready for evaluation	2 h	1 h 30 min	Operator
Task-063: Package and test the project before final deliverable/T1	Make sure the zip project is correct and all the requirements are fulfilled.	30 min	30 min	Tester

Capturas de tablón:

Como se ha comentado anteriormente, al realizar este reporte a una semana de la entrega, simular un tablero inicial no tiene sentido, no obstante, se mostrará el tablero en su situación actual:



Como se puede observar se han finalizado un total de 111 tareas hasta el momento, y como estamos en una etapa final del proyecto, lo único que falta por hacer es realizar el reporte de planificación y progreso y entregar el proyecto.

Presupuesto

ROLES	Coste por hora	Horas estimadas	Coste estimado	Horas reales	Coste real
Manager	30 €	16h 45min	502,50 €	19h 55min	597,50 €
Analista	30 €	7h 50min	235,00 €	8h 20min	250,00 €
Desarrollador	20 €	79h 45min	1.595,00 €	87h 55min	1.758,33 €
Tester	20 €	25h 10min	503,33 €	24h 30min	490,00 €
Operador	20 €	2h 10min	43,33 €	1h 45min	35,00 €
Todos los roles	20 €	61h 20min	1.226,67 €	60h 20min	1.206,67 €
Total		193h 00min	4.105,83 €	202h 45min	4.337,50 €

Para la amortización contamos con dos equipos, un ordenador de sobremesa valorado en 2,000 € y un portátil para los follow-ups valorado en 900 €.

Amortización mensual: $(2000 \text{ €} / 36) + (900 \text{ €} / 36) = 55,56 \text{ €} + 25,00 \text{ €} = 80,56 \text{ €}$

Progreso

A continuación, se comentará el registro de progreso, al ser un solo miembro en el grupo y haberlo realizado todos los requisitos como único integrante, lo enfocaré con un tono de retrospectiva.

Nombre: Gabriel Vacaro

Indicador de rendimiento:

- Historias de usuario completadas: 117
- Asistencia a clases teóricas y prácticas: Asistencia perfecta
- Ratio de commits por historia de usuario: 1,03

Valor alcanzado: Como se indica en el *chartering report*, la nota objetivo de la asignatura en un inicio era de 5.0 puntos, no obstante, al final, debido a la dedicación, el esfuerzo, y los ánimos del profesor Don Rafael Corchuelo, así como los buenos consejos que he recibido a lo largo de este curso, me han impulsado a ir a por la matrícula. Considero que la metodología aplicada ha sido de gran ayuda, y espero que lo siga siendo en mi futura vida profesional, pues la rigurosidad del método me ha permitido expresar al máximo mi capacidad productiva.

Evaluación: Considero que mi desempeño ha sido excelente, así lo reflejan las métricas y las conversaciones con el cliente en los follow-ups.

Recompensa: Durante el transcurso de esta asignatura, he podido observar cómo he ido mejorando en el manejo de soft skills, así como desarrollador, tester, analista y mánager. Esto se puede ver reflejado en como disminuyen los tiempos de implementación en las últimas features, a diferencia de las primeras que implementé. El proceso de aprendizaje fue muy beneficioso, entretenido y gratificante, sin lugar a dudas esa recompensa es suficiente. Por otra parte, tras seis convocatorias sin poder superar la asignatura, cerrar esta etapa con una matrícula sería otra buena recompensa.

Conflictos:

Considero que no ha habido ningún conflicto.

Comparativa entre presupuesto real y estimado

Al comparar el coste estimado del proyecto con el coste real una vez finalizado, se observa una diferencia total de 231,67 € (un incremento aproximado del 5,64% sobre el presupuesto inicial). Esta desviación es relativamente moderada y puede considerarse aceptable en el contexto de un proyecto académico con múltiples roles asumidos por una sola persona.

El principal origen de este incremento se encuentra en tres áreas clave:

- Mayor dedicación en el rol de Manager, que aumentó en más de 3 horas respecto a lo previsto. Esto puede deberse a que al ser la primera vez que se sigue esta metodología de forma rigurosa, se ha subestimado el tiempo necesitado.
- Incremento de horas en el rol de Desarrollador, con más de 8 horas adicionales, lo cual si bien es sustancial, considero que para la experiencia que poseo, la estimación ha sido bastante acertada y refleja que el rol con una mayor cantidad invertida de horas tampoco se desvía en gran medida.
- Ligera desviación en el rol de Analista, debido al tiempo extra invertido en la elaboración del informe de análisis global.

Por otro lado, algunos roles como Tester y Operador han requerido algo menos de tiempo del previsto, compensando parcialmente el exceso en los otros roles. En el caso de las tareas realizadas por todos los roles, la estimación ha sido precisa, pues viene dictaminada por el horario de clases en su mayor parte.

En resumen, la desviación es asumible y responde a un ajuste natural del esfuerzo requerido en la recta final del proyecto, lo cual se traduce en un entregable más robusto y completo. Esta experiencia pone de manifiesto la importancia de mantener márgenes de seguridad en la planificación inicial y la utilidad de un seguimiento continuo del esfuerzo invertido por rol.

Conclusiones

Este informe ha permitido reflejar de forma clara y estructurada tanto la planificación como el progreso alcanzado a lo largo del desarrollo del proyecto. A través de un análisis detallado de las tareas realizadas, la evolución del trabajo y la gestión del tiempo y los recursos, se evidencia un compromiso constante con la calidad, la organización y el cumplimiento de los objetivos.

Pese a una ligera desviación presupuestaria respecto a las estimaciones iniciales, el esfuerzo adicional se ha traducido en un entregable sólido y completo, acorde con los requisitos del curso. La gestión unipersonal del proyecto ha supuesto un reto añadido, pero también ha facilitado una visión global y una mayor responsabilidad sobre cada aspecto de este.

En conjunto, el trabajo realizado pone de manifiesto una evolución continua, tanto a nivel técnico como metodológico, y sienta unas bases sólidas para afrontar futuros proyectos con una mayor madurez y eficiencia.

Bibliografía

Intentionally blank.